॥ वसुधैव कुटुम्बकम् ॥

SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act 1956)

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Semester: III **Year:** AY 26-27

Subject Title: Operating Systems Lab	
EXPERIMENT No: 5	
TITLE: Conditional Statements	**DoP**: 10-08-2026

Aim: Implement shell script to demonstrate conditional statements.

Learning Outcome: 1. To understand the conditional statements
2. To demonstrate shell script to implement conditional statements using Linux Commands

Hardware/Software:

This experiment is performed using Ubuntu OS in a virtualized environment within Windows 11. The software used is:

- VMware Workstation 17 Player (Free Virtualization Tool)
- Ubuntu 22.04.1 LTS ISO file
- VMware Tools installed on the virtual machine

Problem Statement:

A program should be written where all the following conditions should be satisfied.

1. Shell script to print whether the number entered by the user is even or odd
2. Shell script to print the largest of three numbers
3. Shell script to print whether the year entered by the user is a leap year or not
4. Shell script to calculate the balance of the account based on the following conditions:
    a. Accept the account balance from the user
    b. Accept withdrawal amount from the user
    c. If the withdrawal amount is <1500, calculate the tax as 3% of the withdrawal amount
    d. If the withdrawal amount > 1500 and less than 3000, calculate the tax as 4% of the withdrawal amount
    e. If the withdrawal amount > 3000 then calculate the tax as 5% of the withdrawal

amount

f. If the balance is less than the withdrawal amount, print insufficient balance
g. Print the amount withdrawn along with the tax deducted

Theory:

The shell acts as an interface between the user and the OS services. It accepts human-readable commands from the user and converts them into a language that the kernel can understand. The shell can be accessed using the terminal.

Shell accepts the command as input from the user and executes it. A shell can also take inputs as a file and execute them in case of executing a bunch of commands routinely. This helps to avoid the repetitive work of typing each command in the terminal. This file containing a bunch of commands is known as a shell script and is saved with the extension of .sh.

There are five types of conditional statements which can be used in shell scripting. These include:

1. **if statement**
   processes the statements in the if block, when the condition specified, is true
   Syntax:

```
if [condition]
then
  statements
fi
```

2. **if-else statement**
   If the specified condition is false, then the statements in the else block will be executed
   Syntax:

```
if [condition]
then
  statements-block-1
else
  statements-block-2
fi
```

3. **Else if Ladder → if .. elif .. else..fi statements**
   To use multiple conditions in one if-else block, the elif keyword is used in the shell. If the condition1 is true then it executes statement-block-1, and this process continues for each elif statement. If none of the conditions is true then it processes the else part.
   Syntax:

```
if [condition1]
then
  statements-block-1
elif[condition2]
then
  statements-block-2
else
  statements-block-3
fi
```

4. **Nested if Ladder → if..then..else..if..then..fi..fi.. statements**
   A nested if-else block can be used when one condition is satisfied then it again checks another condition. In the syntax, if expression1 is false then it processes the else part, and again expression2 will be checked.
   Syntax:

```
if [expression1]
then
  statement1
  statement2
  .
else
  if [expression2]
  then
    statement3
    .
  fi
fi
```

5. **Switch Statement**
   case statement works as a switch statement if the specified value match the pattern then it will execute a block of that particular pattern
   When a match is found all of the associated statements until the double semicolon (;;) is executed.
   A case will be terminated when the last command is executed.
   If there is no match, the exit status of the case is zero.
   Syntax:

```
case  in
  Pattern 1) Statement 1;;
  Pattern n) Statement n;;
esac
```

Command	***Description***
echo	Display the line of text/string passed to it as an argument
read	Read the value passed and assign it to the variable
expr	Evaluate the expression provided and display the result as output
-eq	Check if the value of two operands is equal or not
-ge	Checks if the value of the left operand is greater than or equal to the value of the right operand
-gt	Checks if the value of the left operand is greater than the value of the right operand
-ne	Checks if the value of two operands is not equal
exit	exit a shell with a given status.
-lt	Checks if the value of the left operand is less than the value of the right operand
scale	Specifies the number of digits after decimal place

Note:
By default, the .sh file may not have executable permission. Hence, using the command chmod +x filename, the file is given executable permission.

#!/bin/sh signifies bourne shell compatible script

Algorithm:
For performing conditional statements using shell scripting:

1. Take input from the user and store it in some variables
2. Specify the condition to be evaluated as an if block
3. Use the if-else statement to evaluate a condition and execute different commands based on the outcome of the condition
4. Use the elif statement to evaluate multiple conditions and execute different commands based on the outcome of each condition.
5. Print the output

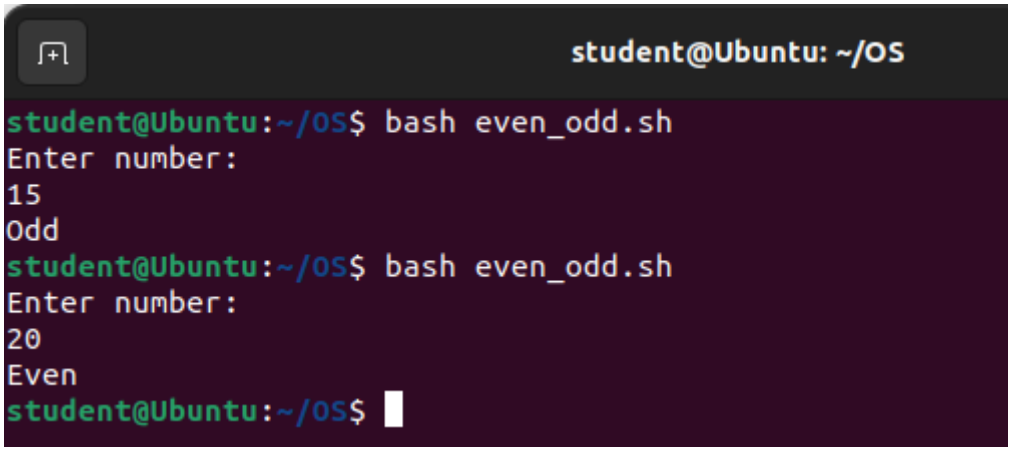
1. Program:

Open

even_odd.sh
~/OS

```
1 #! /bin/sh
2
3 echo "Enter number:"
4 read n
5
6 if [ $((n % 2)) -eq 0 ]
7 then
8     echo "Even"
9 else
10     echo "Odd"
11 fi
12
```

Output Screenshots:



2. Program:

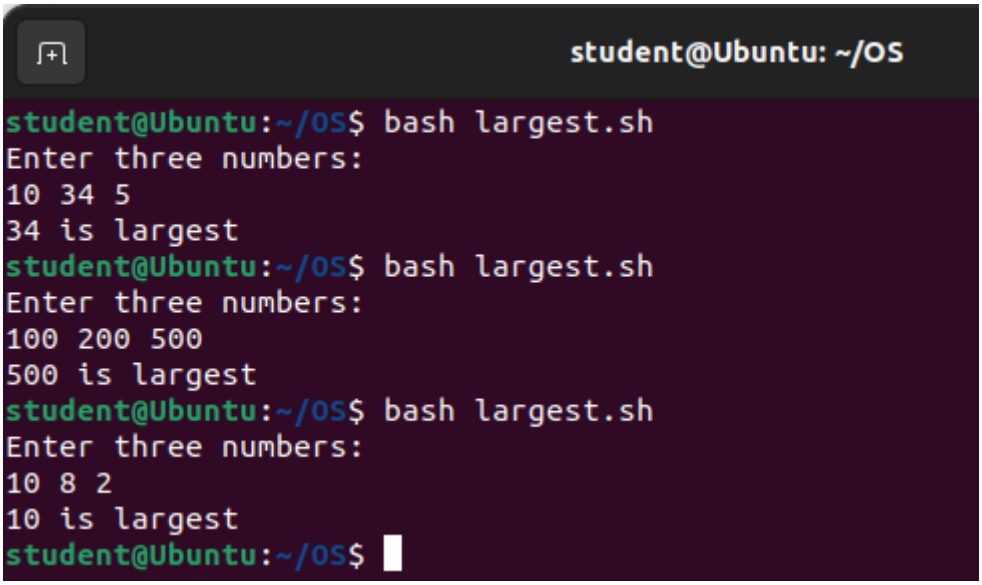
Open

largest.sh
~/OS

largest.sh

```
1 #! /bin/sh
2
3 echo "Enter three numbers:"
4 read a b c
5
6 if [ $a -gt $b ] && [ $a -gt $c ]
7 then
8     echo "$a is largest"
9 elif [ $b -gt $c ]
10 then
11     echo "$b is largest"
12 else
13     echo "$c is largest"
14 fi
15
```

Output Screenshots:



3. Program:

Open

leap_year.sh
~/OS

leap_year.sh | largest.sh

```
#! /bin/sh

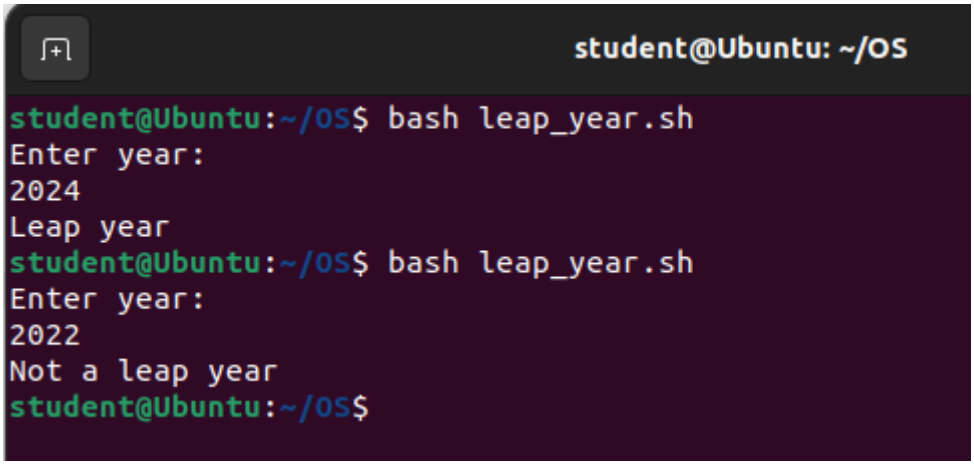
echo "Enter year:"
read y

x=`expr $y % 4`

if [ $x -eq 0 ]
then
    echo "Leap year"
else
    echo "Not a leap year"
fi

```

Output Screenshots:



4. **Program:**

Open

account.sh
~/OS

account.sh | leap_year.sh

```
#!/bin/bash

echo "Enter account balance:"
read balance

echo "Enter withdrawal amount:"
read withdrawal

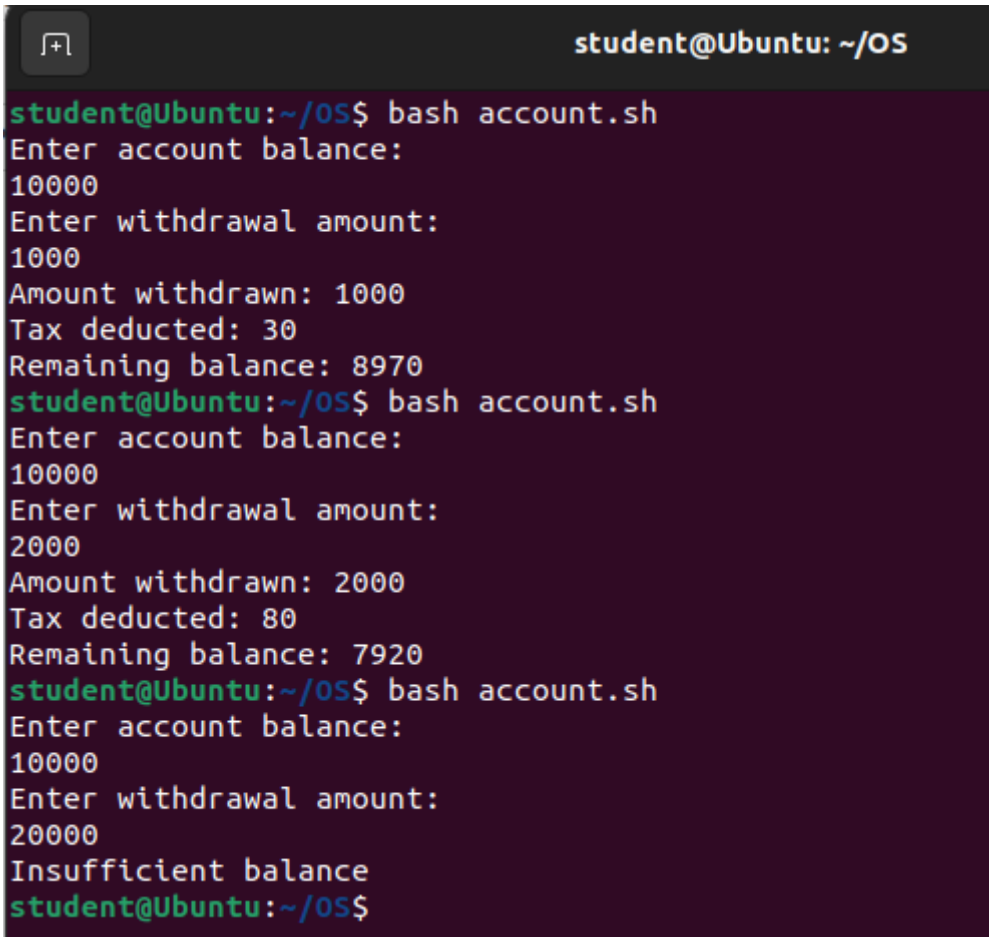
if [ $balance -lt $withdrawal ]
then
    echo "Insufficient balance"
else
    if [ $withdrawal -lt 1500 ]
    then
        tax=$((withdrawal * 3 / 100))
    elif [ $withdrawal -gt 1500 ] && [ $withdrawal -lt 3000 ]
    then
        tax=$((withdrawal * 4 / 100))
    elif [ $withdrawal -gt 3000 ]
    then
        tax=$((withdrawal * 5 / 100))
    else
        tax=0
    fi

    echo "Amount withdrawn: $withdrawal"
    echo "Tax deducted: $tax"

    balance=$((balance - withdrawal - tax))

    echo "Remaining balance: $balance"
fi
```

Output Screenshots:



Conclusion:

The experiment on conditional statements using Linux Commands has been successfully completed. It helped to understand the capabilities of a shell script for performing various conditional statements. The experiment helped to learn the commands and syntax involved in performing different conditional constructs.

It shows the integration of various conditional structures such as if, if-else, if-elif-else, and other shell scripting commands in Linux to automate expression evaluation based on dynamic user input.

Assessment Schemes:

Attendance	Discipline	Short oral	Correctness of Lab Report	Timely completion of Lab Report	Total marks (10)	Signature of Teacher with Date